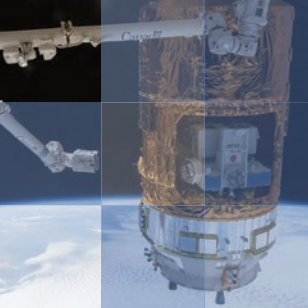


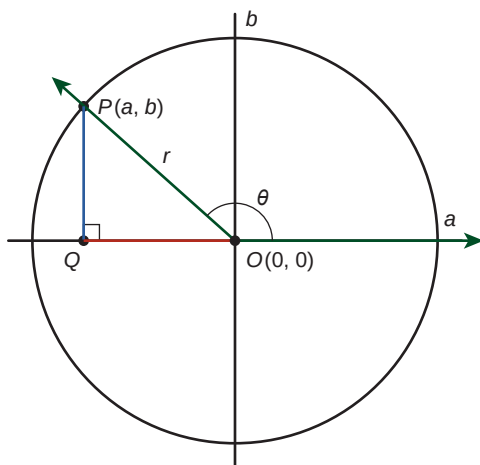


Trigonometric Functions of Any Angle

Trigonometric functions are defined for angles of any measure.

Defining Trigonometric Functions for Angles on Circles with Radius r

You can determine the trigonometric functions of an angle θ given a point (a, b) on its terminal side, even if the point does not lie on the unit circle. The circle shown has radius r . The angle θ is in standard position, and the point $P(a, b)$ lies on its terminal side. A segment from P to the horizontal axis at point Q creates right triangle PQO with the origin.



The distance r between points $P(a, b)$ and $O(0, 0)$ can be determined using the distance formula.

$$r = \sqrt{(a - 0)^2 + (b - 0)^2} = \sqrt{a^2 + b^2}$$

Then the trigonometric functions of θ are determined by the side lengths of triangle PQO .

Definition

Let (a, b) be a point other than the origin on the terminal side of an angle θ in standard position. The distance from the point to the origin is

$$r = \sqrt{a^2 + b^2}.$$

The **trigonometric functions** of θ are defined as follows:

$$\sin \theta = \frac{b}{r} \quad \cos \theta = \frac{a}{r} \quad \tan \theta = \frac{b}{a} \quad (a \neq 0)$$

► **Think About It** The point (a, b) cannot be the origin because that leads to a value of r equal to 0. In that case, the values of sine and cosine have 0 in the denominator and are therefore undefined. If only the horizontal coordinate is 0, then $\tan \theta$ is undefined.

EXAMPLE 1

Let $(-4, -11)$ be on the terminal side of an angle θ . Determine the exact value of the trigonometric ratio.

A $\sin \theta$

SOLUTION

Determine r .

$$r = \sqrt{(-4)^2 + (-11)^2} = \sqrt{16 + 121} = \sqrt{137}$$

$$\sin \theta = \frac{b}{r} = \frac{-11}{\sqrt{137}} = \frac{-11\sqrt{137}}{137}$$

B $\cos \theta$

SOLUTION

$$\cos \theta = \frac{a}{r} = \frac{-4}{\sqrt{137}} = \frac{-4\sqrt{137}}{137}$$

C $\tan \theta$

SOLUTION

$$\tan \theta = \frac{b}{a} = \frac{-11}{-4} = \frac{11}{4} \quad \blacksquare$$

► **Think About It** Pay attention to the sign of each horizontal and vertical coordinate. While the value of r does not depend on the signs, the function values do.

Using the Equation of a Line

If the terminal side of an angle x is given as an equation of a line, determine the trigonometric functions of the angle by graphing the line and then determining a point on the line.

► **Think About It** It is often customary to use a variable such as x from the English alphabet for an angle measured in radians and a variable such as θ from the Greek alphabet for an angle measured in degrees.

EXAMPLE 2

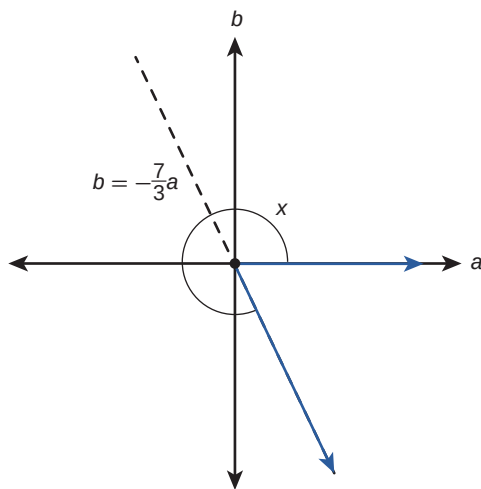
The terminal side of an angle x in standard position coincides with the line $7a + 3b = 0$ in Quadrant IV. Determine $\cos x$.

SOLUTION

Step 1 Solve the equation for b .

$$\begin{aligned}7a + 3b &= 0 \\3b &= -7a \\b &= -\frac{7}{3}a\end{aligned}$$

Step 2 Graph the angle with its terminal side in Quadrant IV.



Step 3 In Quadrant IV a is positive. Choose a positive value of a to determine the coordinates of a point (a, b) on the terminal side of x .

When $a = 3$, $b = -\frac{7}{3}a = -\frac{7}{3}(3) = -7$.

The point $(3, -7)$ is on the terminal side of the angle.

Step 4 Determine r .

$$r = \sqrt{3^2 + (-7)^2} = \sqrt{9 + 49} = \sqrt{58}$$

Step 5 Use the definition of $\cos x$.

$$\cos x = \frac{a}{r} = \frac{3}{\sqrt{58}} = \frac{3\sqrt{58}}{58} \quad \blacksquare$$

Using Reference Angles

Every nonquadrantal angle in standard position is associated with a reference angle.

Definition

A **reference angle** x' for an acute angle x is the positive acute angle made by the terminal side of x and the horizontal axis.

To determine the reference angle for an angle x , draw the angle in standard position, and then draw the acute angle coinciding with x that has its initial or terminal side on the horizontal axis.

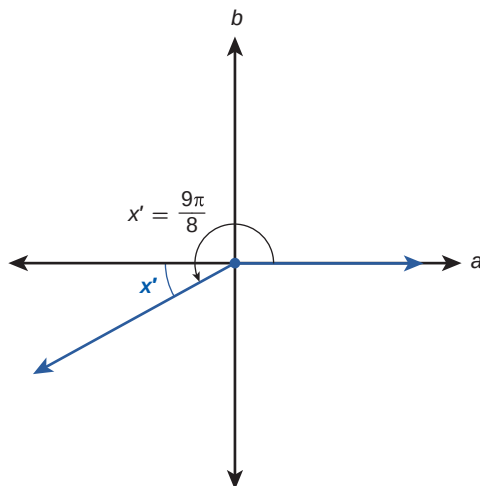
► **Think About It** If an angle measure is greater than 2π or 360° , first determine the measure of an angle that is coterminal with the given angle, and then determine its reference angle.

EXAMPLE 3

Determine the measure of the reference angle for $\frac{9\pi}{8}$.

SOLUTION

Graph the angle.



- **Remember** State the measure of an angle in the units given in the problem unless otherwise directed.

The measure of the reference angle x' is the difference between $\frac{9\pi}{8}$ and the angle π made by the negative horizontal axis.

$$x' = \frac{9\pi}{8} - \pi = \frac{9\pi}{8} - \frac{8\pi}{8} = \frac{\pi}{8} \quad \blacksquare$$

► **Q&A**

Q What is the measure of $\frac{9\pi}{8}$ in degrees?

A 202.5°